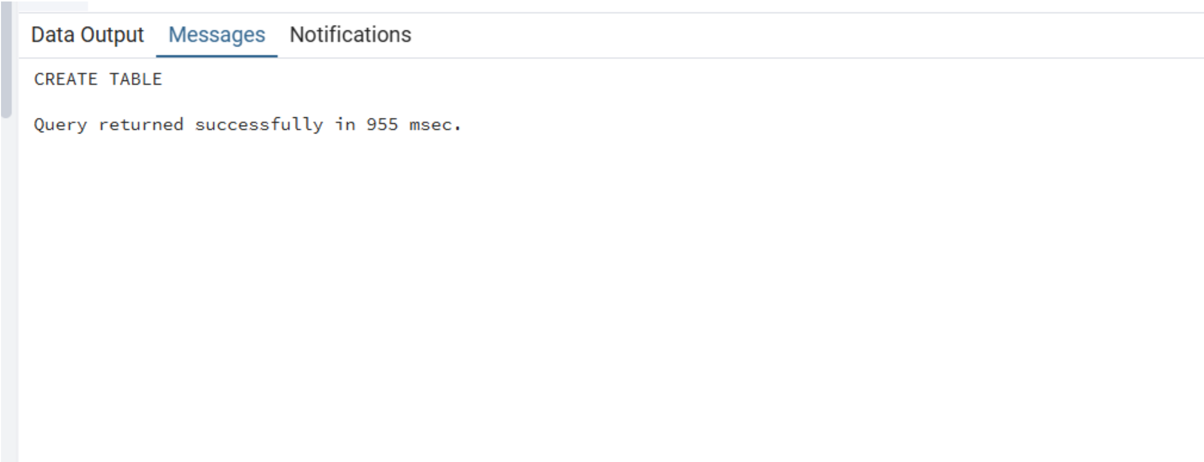
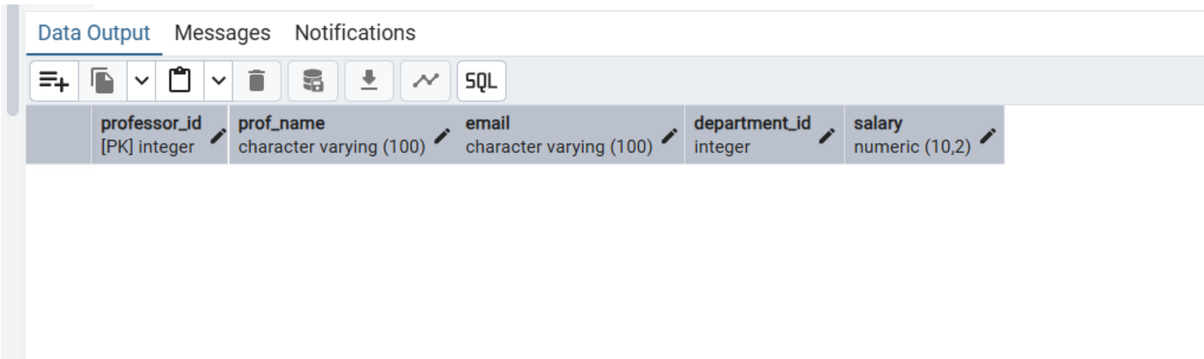
HANDS-ON 1 [Beginner] Schema Design & Core SQL — DDL and Normalisation

Task 1: Create the Database and Tables

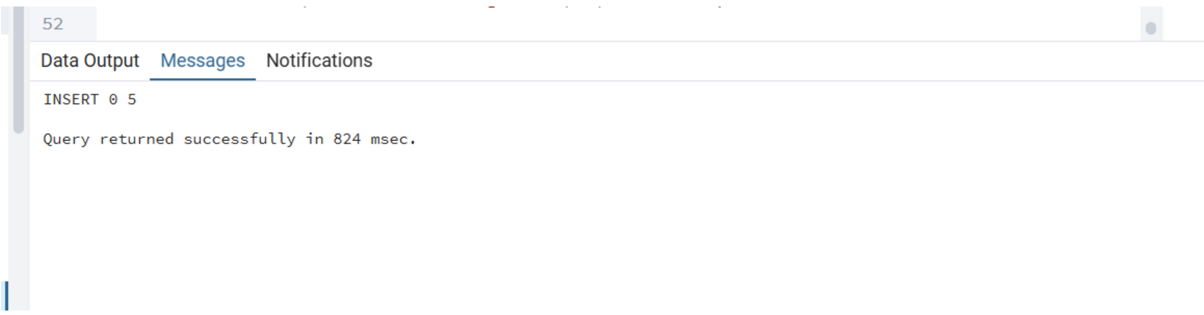
1.Table Created



2. Retrieve empty tables as no data has been inserted yet



3. Insert sample data into the tables



4. Check if all tables have been populated with data—Count of Courses has been retrieved

	count bigint
1	5

5.Retrieve all data from the tables to verify the inserted records—Table Professor has been retrieved

Showing rows: 1 to 5 Page No: 1

	professor_id [PK] integer	prof_name character varying (100)	email character varying (100)	department_id integer	salary numeric (10,2)
1	1	Dr. Anand Krishnan	anand.k@college.edu	1	95000.00
2	2	Dr. Meena Pillai	meena.p@college.edu	1	88000.00
3	3	Dr. Sunil Rajan	sunil.r@college.edu	2	82000.00
4	4	Dr. Latha Gopal	latha.g@college.edu	3	79000.00
5	5	Dr. Kartik Bose	kartik.b@college.edu	4	76000.00

Task 2: Verify Normalisation

```
-- TASK 02:Normalization Analysis
-- 1NF (First Normal Form)
-- All tables contain atomic (single) values in every column.
-- There are no repeating groups or multiple values stored in one field.
-- For example, storing multiple phone numbers in one column like
'9876543210,9123456789'
-- would violate 1NF. Instead, each phone number should be stored separately.

-- 2NF (Second Normal Form)
-- Every non-key attribute is fully dependent on the primary key.
-- In the enrollments table, the logical candidate key is (student_id,
course_id).
-- The attribute grade depends on the complete student-course enrollment,
-- not on student_id or course_id individually. Hence the table satisfies 2NF.

-- 3NF (Third Normal Form)
-- There are no transitive dependencies in the schema.
```

```
-- Student details do not store department names directly.
-- Instead, students reference department_id, and department details are
stored
-- only in the departments table.
-- Therefore, the schema satisfies Third Normal Form (3NF).
```

Task 3: Alter and Extend the Schema

1.Add Phone Number to Students Table

Data Output | Messages | Notifications

```
ALTER TABLE

Query returned successfully in 275 msec.
```

2.verify the addition of new column

Data Output | Messages | Notifications

	column_name name
1	student_id
2	first_name
3	last_name
4	email
5	date_of_birth
6	department_id
7	enrollment_year
8	phone_number

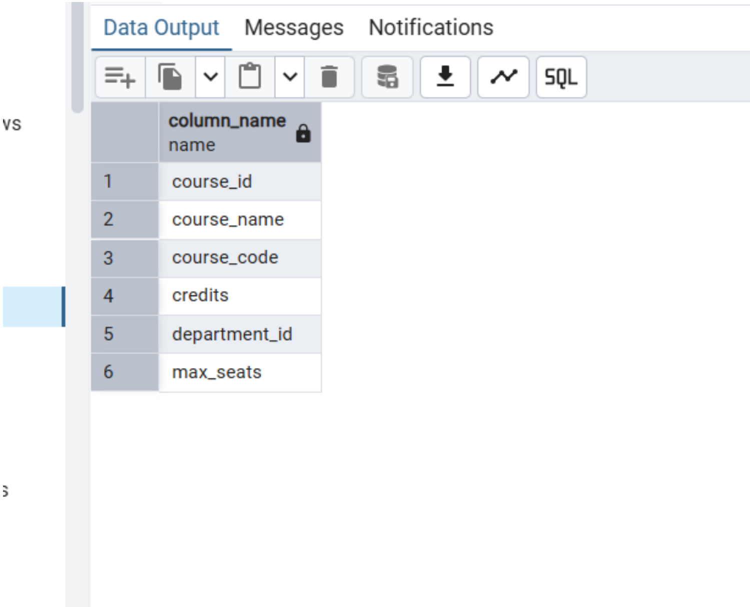
3. Add max_seats to Courses Table

Data Output | Messages | Notifications

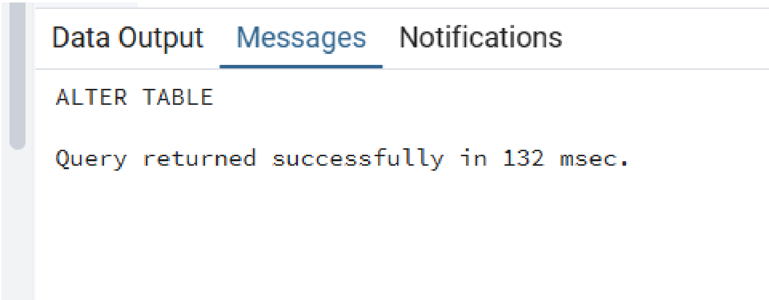
```
ALTER TABLE

Query returned successfully in 225 msec.
```

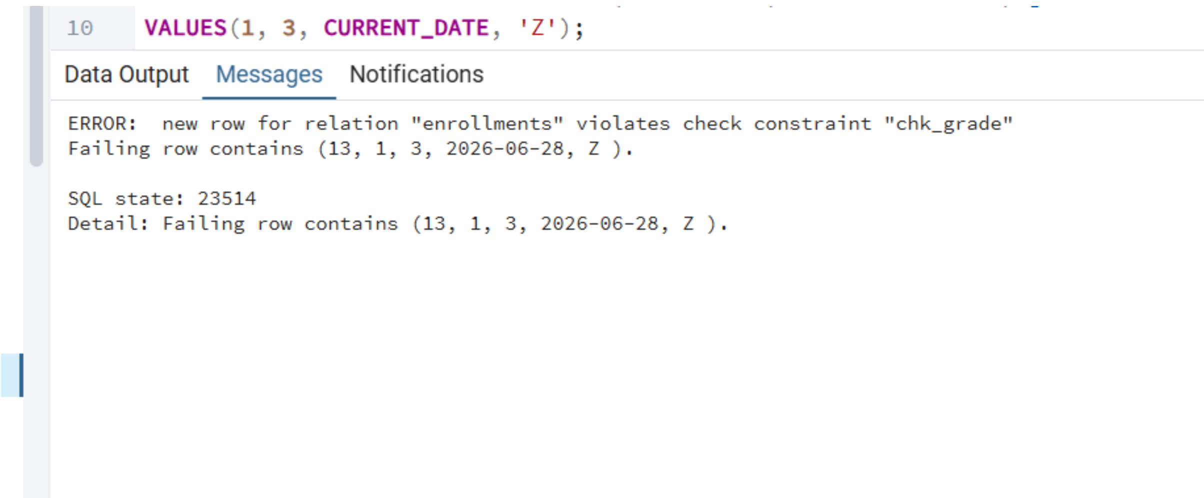
4. Verify the addition of the new column



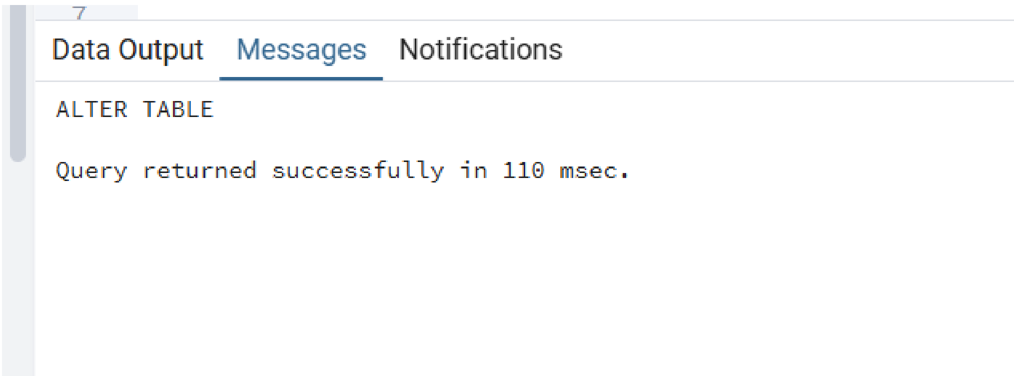
5. Add CHECK Constraints on grade



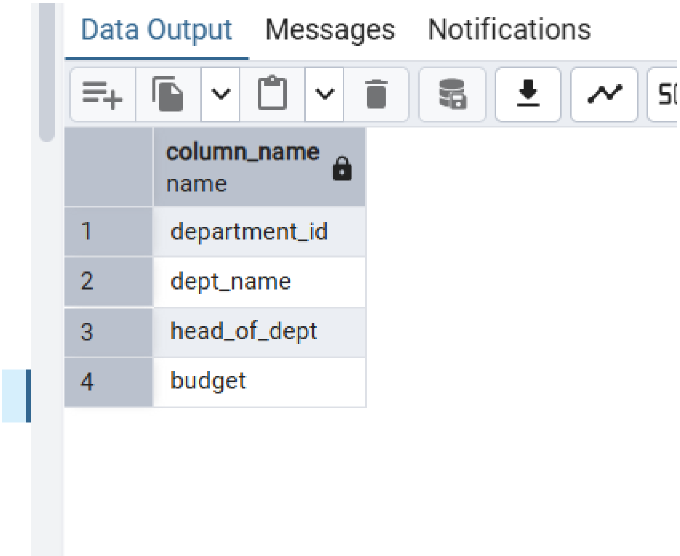
6. Verify the CHECK constraint



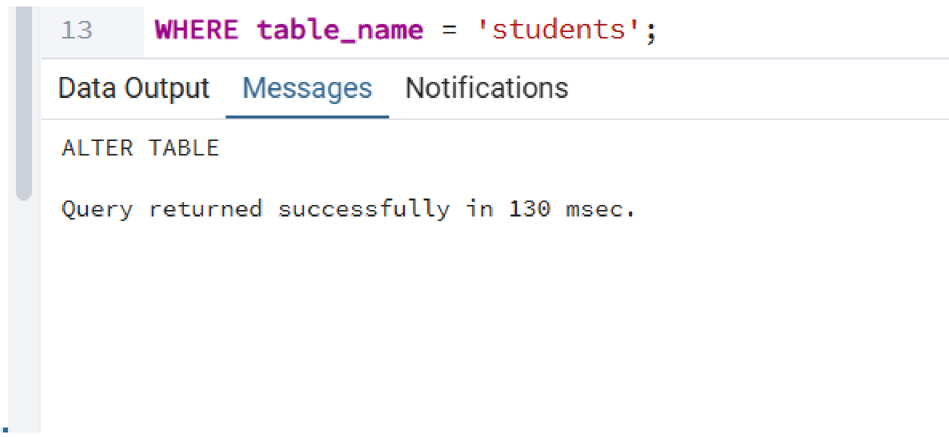
7. **Rename hod_name to head_of_dept**



8. **Verify the renaming of the column**



9. **Rollback the addition of the phone_number column in students table**



10. **Verify the removal of the column**

```sql
SELECT column_name FROM information_schema.c
WHERE table_name = 'students';
```

Data Output Messages Notifications

SQL

	column_name name
1	student_id
2	first_name
3	last_name
4	email
5	date_of_birth
6	department_id
7	enrollment_year